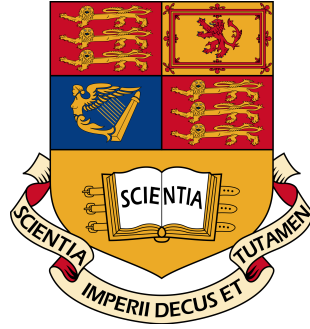




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Project Plan

A Conversational Multi-Agent System for Smart Campus Energy Management

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Abstract

This report proposes the development of a conversational campus energy management system, powered by Large Language Model (LLM)-based agents. It leverages emerging frameworks in Microsoft's AI Agent Services to develop a Multi-Agent System (MAS) capable of real-time monitoring, dynamic decision-making, and user interaction.

The system will include two core mechanisms: (1) a baseline energy management system that continuously analyses energy data. It will support smart prognostics and decision-making to optimise efficiency and costs. (2) a user-facing chatbot that enables interactive queries. The chatbot either responds to system-related queries or adapts the existing settings to match user requests. Finally, I simulate the smart campus environment using mock system log data.

The work aims to demonstrate the feasibility of LLM-based agents for energy systems traditionally dominated by rule-based methods or traditional machine learning models [15, 26, 29, 7]. It aims to highlight the potential for a scalable, interpretable and flexible energy management solution. The system will be benchmarked through a range of curated query tasks and a general user survey (feedback).

1 Introduction

1.1 Problem Description and Motivation

University campuses include a range of spaces (e.g., lecture halls, labs, dining halls) all with distinct energy profiles and usage patterns. This complexity requires tailored monitoring to manage HVAC (heating, ventilation, air-conditioning), lighting, and equipment efficiently [20]. There are a range of factors to consider when optimising energy usage (e.g., grid prices, weather and occupancy [19], utilities [30], user satisfaction [13]). This makes centralised control difficult. There is a need for efficient resource allocation, prognostics, and dynamic decision-making; a smart energy management system.

Artificial intelligence (AI) enhanced multi-agent systems (MAS) have received a lot of attention for such systems. MAS are a distributed AI approach where multiple autonomous agents collaborate or compete to solve complex problems [3]. Each autonomous agent is typically assigned control of different parts of the campus (buildings [26], departments, microgrids [3, 11], etc.). These systems usually adopt rule-based methods enhanced by AI techniques (e.g., ML [12, 4], neural networks [29, 22, 19], reinforcement learning [30, 5, 1]). MAS approaches are attractive; they naturally align with the distributed nature of campus energy (e.g., each building is an agent). However, traditional MAS systems struggle to interpret and dynamically adapt to users' preferences. Adjusting system preferences requires high maintenance skills (e.g., re-training the AI models) [4]. Moreover, there is a lack of standardised communication protocols, and frameworks [21, 4]. This makes it difficult to adapt to existing systems when developing new ones. Finally, integrated AI algorithms still face significant obstacles in meeting multiple divergent objectives (e.g., volatile market prices, personalised user requests) [5].

An exciting new trend is incorporating the use of large language models (LLM), to offer more flexible, interpretable, and user-centric alternatives. There are already some recent works exploring LLM-based power grid fault analysis [23]. Another work focused on encoding expert knowledge into control agents [28]. This would give a high-level control to the manager and reduce reliance on system maintenance. The newly emerging LLM-based agent frameworks

(e.g., Semantic Kernel [18], AutoGen [27], Model Context Protocol [2], Azure AI Foundry [17]) are still underexplored and hold great potential. They can help solve a range of issues faced with traditional MAS systems. To name a few; complex user query handling [18], standardised agent communication [2], rapid prototyping [27], accelerated development [17]. Finally, LLMs themselves are an emerging technology and show a lot of promise for the future. An LLM-based multi-agent system is likely to scale and improve with future advances in this field.

1.2 Objectives

The first objective is to develop a baseline energy management system. This system will continuously analyse dynamically updated logs (energy price, weather, occupancy, utility activity, system settings, etc.). It will provide prognostics and decisions to optimise energy efficiency and costs. The second objective is to integrate an LLM-based user-facing chatbot into the baseline system. The chatbot is used to interpret complex user and manager queries. This information will then be used to adapt system preferences, enhancing future analysis and decisions and improving user satisfaction. Finally, on a technical level, the second objective aims to explore the potential of emerging (and still underexplored) LLM-based agent frameworks. These frameworks are discussed in more detail in the methodology.

2 Methodology

2.1 Mock Data Design

The dynamically updated system logs (from sensors, utilities, and energy price changes) are a key component of the first objective (baseline system). Hence the first stage involves designing and generating synthetic data for the system. Mocking gives us a broad range of data, that would otherwise be limited to the existing infrastructure (e.g., available sensors). This allows for faster prototyping and testing. Synthetic data also lets us simulate different energy usage scenarios, testing how well the system generalises.

Using existing monitoring data will help with database design, structure, and data integrity (accuracy and validity) of our synthetic data. Hence, extensive requests were made to existing energy monitoring projects. This includes; Energy Futures Lab at Imperial College [6] and data used by a smart campus case study [9]. I also considered the publicly available UNICON Energy Consumption Dataset [25]. Note that this stage is still ongoing, meaning the mock data structure is not completely finalised.

The general structure captures a range of data types. This includes; real-time and past logs; (1) System logs (such as; grid energy prices, and energy usage), (2) Sensor logs (e.g., building occupancy, lighting levels, indoor temperature), see Figure 1 (3) Utility logs (e.g., HVAC, lighting), (4) Energy settings (system preferences by users or system manager).

The project will use Azure Cosmos Database [16] to store and retrieve data. Cosmos DB is a globally distributed, multi-model NoSQL database. Its flexibility and seamless integration with other used Azure services make it well-suited for handling diverse data logs.

SensorLog	
+ sensorLogId (primary key)	"id": "sensorlog_20250606_0456",
+ buildingId (foreign key)	"buildingId": "building_A",
+ timestamp : datetime	"timestamp":
+ environment data	"2025-06-06T09:10:00Z",
+ temperature	"environment": {
+ humidity	"temperature": 22.8,
+ airQuality	"humidity": 48.1,
+ lighting	"airQuality": 78,
+ occupancy	"lighting": 620,
	"occupancy": 12
	}
	}

Figure 1: An example entity and a JSON sample.

2.2 Multi-Agent Architecture

The proposed system follows a multi-agent architecture, divided into four key layers: interface, decision, control, and mock data. Each layer can contain one or more agents. The interface layer contains a user-facing chatbot and visualisation dashboard. The decision layer handles the analysis and prognostics (estimate lifespan, failures, maintenance). The control layer addresses smart management adjustments. Finally, the data layer manages the mock data (data integrity, communication protocols). Figure 2 shows the general architecture design, with interactions between the layers. This architecture was chosen for its modularity, which improves code maintainability, profiling, and debugging. The decision, control and data layers were largely inspired by existing multi-agent systems. The key addition is the interface layer, supporting an energy monitoring dashboard and a chatbot for user interaction.

The project leverages Azure AI Agent Services to build and deploy the system. Semantic Kernel [18] and AutoGen [27] will be used to develop LLM-based agents orchestrated in a multi-agent system. Model Context Protocol (MCP) [2] will facilitate structured communication among agents. These frameworks were chosen to enable flexible and intelligent agent behaviours, beyond those seen in traditional rule-based systems. They also allow seamless integration with other Microsoft Azure services (e.g., Azure Cosmos DB). They have shown a lot of promise in LLM-based multi-agent systems (e.g., web navigation [10, 14], embodied environments [24], generalist systems [8]). It is interesting to explore their performance in a more complex energy management system. Similarly, to test their integration with traditional AI models in place for such systems.

It should be noted, that these tools are all newly emerging and underexplored. They lack comprehensive proven production robustness in the context of smart energy management. It is therefore one of the project's key objectives to explore the potential of these frameworks.

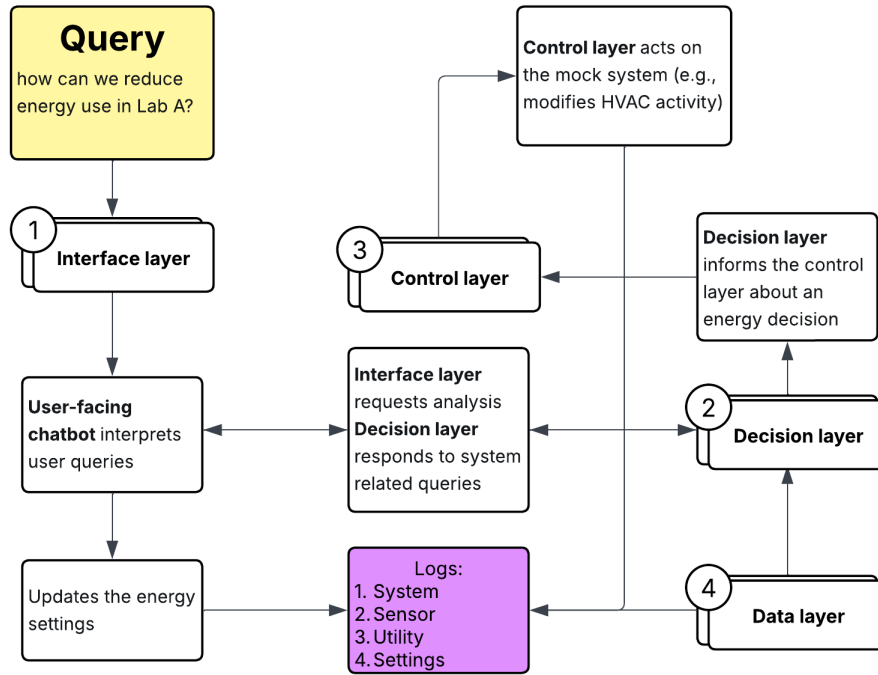


Figure 2: Draft architecture design with 4 layers; interface, decision, control, data.

3 Deliverables and Evaluation

The deliverable for the first objective is a baseline energy management system. This system should have access to and handle dynamically updated logs. It should be able to keep the system within reasonable energy prices and consumption while adhering to existing energy settings (and preferences). The mock data generation and the initial baseline system will be developed in collaboration with James Zhong. I will then focus on the second objective, which is the key focus of this work.

This will be to integrate an LLM-based user-facing chatbot into the system. The chatbot will be used to interpret complex user and manager queries. It will then redirect the requests either to the database (to update preferences) or the decision layer. The interface layer will be tested in two ways; (1) a list of curated queries with a range of task categories (simple commands, analysis requests, security checks, multi-modal input), (2) surveys to collect feedback from the user (students and staff) and system manager. Additionally, the project will include a report that contains a performance analysis and a user guide. Table 1 contains key features of the interface layer, along with their performance metrics.

This benchmarking approach relies on a fixed set of query types, and hence may not fully capture real-world user diversity. It also uses human evaluation for analytical requests and user surveys, which may introduce bias and variability.

Task Type	Example Query	Evaluation Metric
Simple Commands	Stop activity of Utility A	Success rate of action execution (%)
Analytical Requests	How can we reduce energy use in Lab B?	Proportion of actionable suggestions (human-evaluated)
Safety/Security Checks	Set the room temperature to 50°C	Rate of correct safety interventions (%)
Multimodal Queries	Text and voice-based command	Interpretation accuracy across modalities (%)
User Feedback	N/A	Qualitative feedback from survey responses

Table 1: Categories of benchmark tasks for chatbot evaluation, including example queries and performance metrics.

4 Future Plan

4.1 Estimated Timeline

Weeks	Phase	Tasks
Weeks 1–2	Preliminary Research and Drafting	Research existing management solutions and emerging LLM-based agent frameworks, draft a database schema
Weeks 3–4	Data Preparation	Finalise mock data schema, generate all necessary data
Weeks 5–7	Baseline System	Develop a simple baseline system that minimally incorporates all 4 essential layers, gather initial user feedback
Weeks 8–9	Chatbot Benchmark	Create a benchmark for the interface layer; a list of quiz questions and user surveys
Week 10	Interface Layer	Integrate a baseline user-facing chatbot and a monitoring dashboard as part of the interface layer
Weeks 11–14	Extended Features	Integrate and test more of the proposed methodologies into the interface layer
Weeks 15–16	Final Works	Finalise the system, documentation, and the report

Table 2: Project Timeline with phases and tasks

4.2 Progress to Date

The project has completed the initial research and drafting phase. A preliminary literature review has been conducted, identifying past methodologies, and key challenges faced by these methods. A general mock data schema and multi-agent architecture have been designed. The initial project structure has been set up with frameworks to be used in the project (AutoGen, Semantic Kernel, Model Context Protocol).

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